Psychology
UNIVERSITY OF TORONTO

PSY3100H S4 LEC0104:
Machine Learning for Psychological Research

Winter 2022

Mondays, 2 pm – 4 pm , online

Contact Information

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Office Hours: by request

Course Description

In this module I will introduce important concepts for understanding machine learning techniques. We will then work on practical, hands-on exercises for using machine learning methods in psychological research. Students are encouraged to bring data from their own research projects for the applied, hands-on component. A basic understanding of programming in Python is prerequisite.

Objectives

1) To understand important concepts in machine learning.
2) To be able to chose machine learning techniques appropriate for a particular problem.
3) To perform machine learning analyses in Python using existing software libraries.

Reading Material

Textbook: Yuxi (Hayden) Liu (2020). Python Machine Learning By Example, Packt Publishing Ltd., Birmingham, UK. ISBN 978-1-80020-971-8

I use this book as a guide for the class. I find it quite accessible with good practical examples. It is not necessary for the class but may be useful as a reference for your further studies. There are many other good books on the topic and plenty of online resources.

Quercus: Lecture notes, scripts, homework assignments and other important class information will be posted on Quercus. All course assignments should be handed in on Quercus.

Homework Assignments

The class will be evaluated using homework assignments. There will be a homework assignment for each of the six class meetings. Homework assignments are due by the beginning of the next class meeting.

Each homework assignment is worth 20 points. The lowest of the six scores will be dropped from the grade. The maximum score for the class will be 100 points. The course grade will be based on the points but will not necessarily be identical to the points.

Course Policies

Penalties for Lateness
Late submission of homework assignments will be penalized with 1 point deduction per day or fraction of a day.

Religious accommodation
As a student at the University of Toronto, you are part of a diverse community that welcomes and includes students and faculty from a wide range of backgrounds, cultural traditions, and spiritual beliefs. For my part, I will make every reasonable effort to avoid scheduling tests, examinations, or other compulsory activities on religious holy days not captured by statutory holidays. Further to University Policy, if you anticipate being absent from class or missing a major course activity (like a test, or in-class assignment) due to a religious observance, please let me know as early in the course as possible, and with sufficient notice (at least two to three weeks), so that we can work together to make alternate arrangements.

Specific Medical Circumstances
If you become ill and it affects your ability to do your academic work, consult me right away. Normally, I will ask you for medical documentation in support of your specific medical circumstances. The University's Verification of Student Illness or Injury (VOI) form is recommended because it indicates the impact and severity of the illness, while protecting your privacy about the details of the nature of the illness. You can submit a different form (like a letter from the doctor), as long as it is an original document, and it contains the same information as the VOI.

Accommodation for Personal Reasons
There may be times when you are unable to complete course work on time due to non-medical reasons. If you have concerns, speak to me. It is also a very good idea to speak with an advisor in your College Registrars office; they can support you in requesting extensions or accommodations, and importantly, connect you with other resources on campus for help with your situation.

Academic Resources

Ongoing Learning Disability or Accommodation Requirement
Students with diverse learning styles and needs are welcome in this course. If you have an ongoing disability issue or accommodation need, you should register with Accessibility Services (AS) (accessibility.utoronto.ca) at the beginning of the academic year. Without registration, you will not be able to verify your situation with your instructors, and instructors will not be advised about your accommodation needs. AS will then assess your medical situation, develop an accommodation plan with you, and support you in requesting accommodation for your course work. Remember that the process of accommodation is private: AS will not share details of your condition with any instructor, and your instructors will not reveal that you are registered with AS.

Writing:
As a student here at the University of Toronto, you are expected to write well. The university provides its students with a number of resources to help them achieve this. For more information on campus writing centres and writing courses, please visit http://www.writing.utoronto.ca/.

If English is not your first language, you may want to consider the English Language Learning program of the Faculty of Arts and Science:
http://www.artsci.utoronto.ca/current/advising/ell

Academic Integrity and Plagiarism:
Academic integrity is essential to the pursuit of learning and scholarship in a university, and to ensuring that a degree from the University of Toronto is a strong signal of each student's individual academic achievement. As a result, the University treats cases of cheating and plagiarism very seriously. The University of Toronto's Code of Behaviour on Academic Matters (**www.governingcouncil.utoronto.ca/policies/behaveac.htm**) outlines the behaviours that constitute academic dishonesty and the processes for addressing academic offences.

All suspected cases of academic dishonesty will be investigated following procedures outlined in the Code of Behaviour on Academic Matters. If you have questions or concerns about what constitutes appropriate academic behaviour or appropriate research and citation methods, you are expected to seek out additional information on academic integrity from your instructor or from other institutional resources (see **www.utoronto.ca/academicintegrity/resourcesforstudents.html**).

Other Resources
Student Life Programs and Services (http://www.studentlife.utoronto.ca/)
Academic Success Services (http://www.asc.utoronto.ca/)
Counselling and Psychological Services (http://www.caps.utoronto.ca/main.htm)

Provisional Course Outline *(subject to change)*

Date	Topic
Jan. 24th	Overview; first steps in Google Collab; data pre-processing
Jan. 31st	Classifiers; evaluating classifier performance
Feb 14th	Bias/variance trade-off; overfitting/underfitting; combining classifiers
Feb 28th	Regression techniques
March 14th	Convolutional Neural Networks
March 28th	Clustering; topic models; reinforcement learning